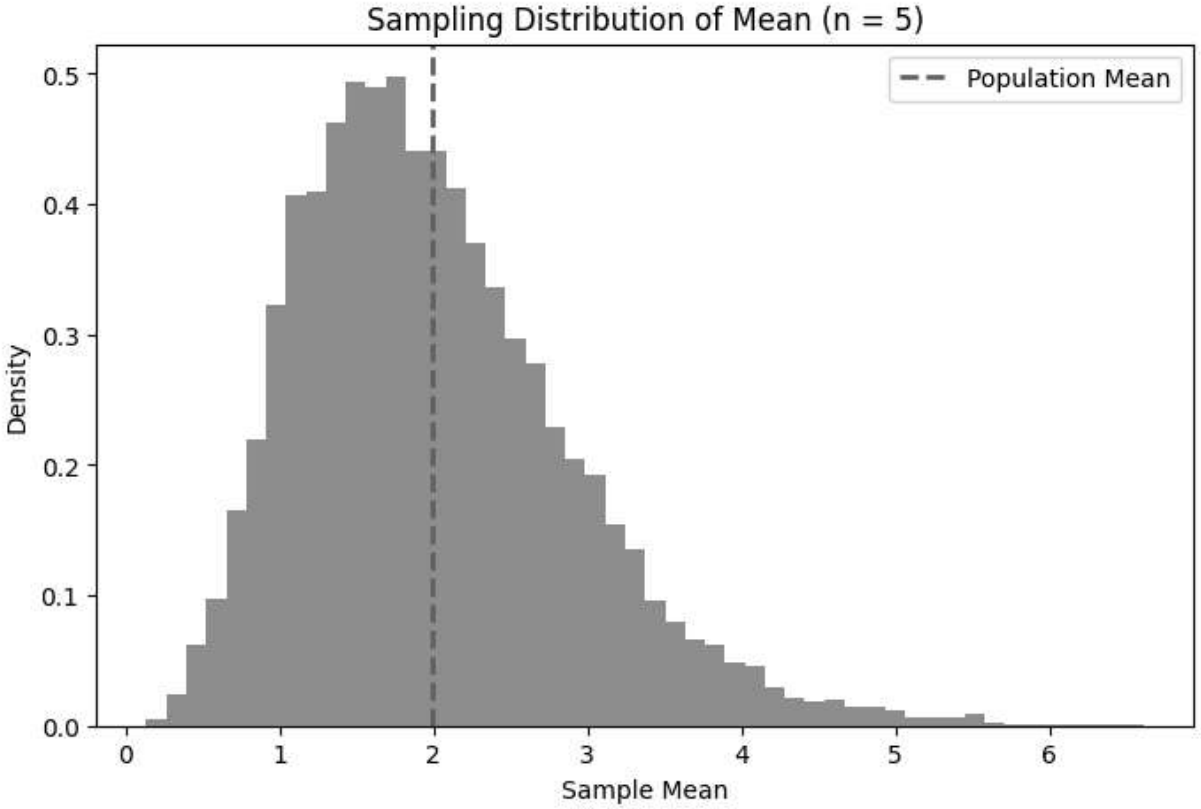
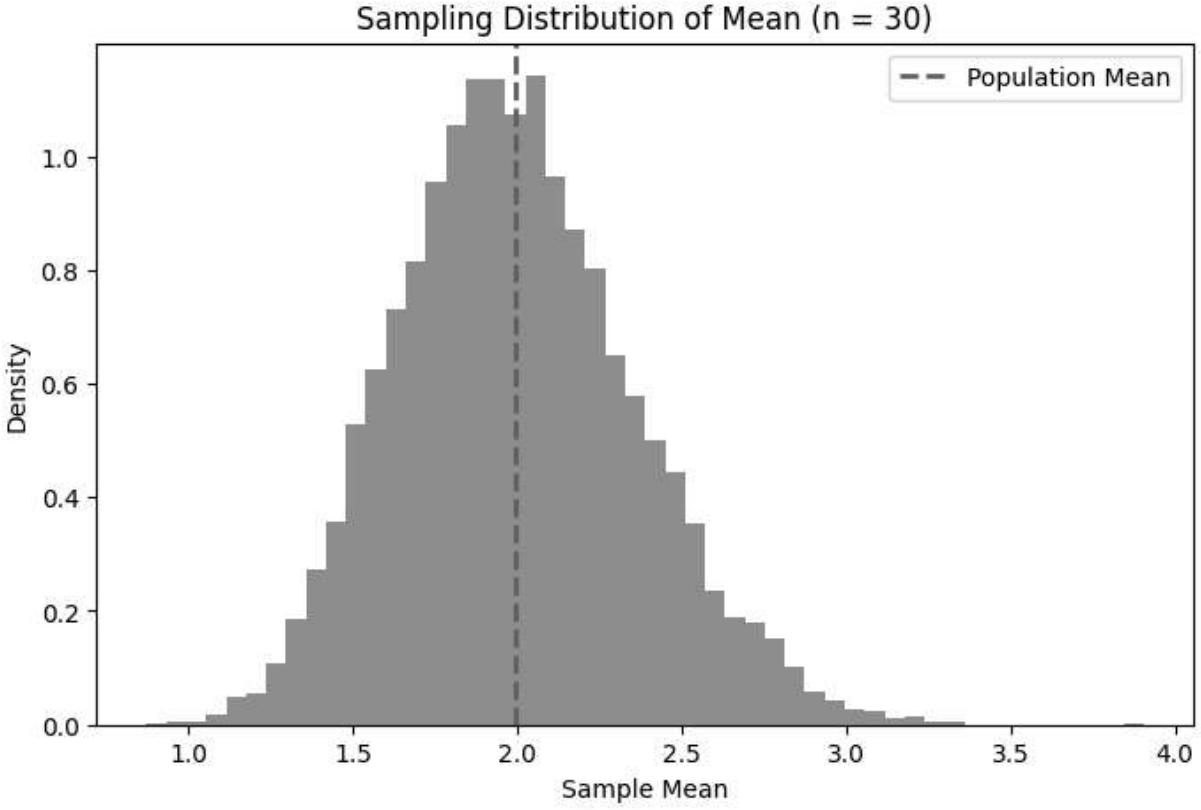
In [5]:
```
# ============================================================
# EXPERIMENT: DEMONSTRATION OF CENTRAL LIMIT THEOREM
# USING SIMULATIONS
# ============================================================
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
# ------------------------------------------------------------
# STEP 1: Set random seed
# ------------------------------------------------------------
np.random.seed(42)
# ------------------------------------------------------------
# STEP 2: Define population
# Exponential distribution: mean = 2, std = 2
# ------------------------------------------------------------
population_size = 100000
population = np.random.exponential(
    scale=2,
    size=population_size
)
population_mean = population.mean()
population_std = population.std()
print("Population Mean:", population_mean)
print("Population Standard Deviation:", population_std)
# ------------------------------------------------------------
# STEP 3: Simulation parameters
# ------------------------------------------------------------
num_simulations = 10000
sample_sizes = [5, 30, 100]
results = []
# ------------------------------------------------------------
# STEP 4: Generate sampling distributions
# ------------------------------------------------------------
for n in sample_sizes:
    sample_means = []
    for i in range(num_simulations):
        sample = np.random.choice(
            population,
            size=n,
            replace=True
        )
        sample_mean = np.mean(sample)
        sample_means.append(sample_mean)
    sample_means = np.array(sample_means)
    simulated_mean = np.mean(sample_means)
    simulated_std = np.std(sample_means)
    theoretical_se = population_std / np.sqrt(n)
    results.append({
        "Sample_Size": n,
        "Simulated_Mean": simulated_mean,
        "Simulated_SE": simulated_std,
        "Theoretical_SE": theoretical_se
    })
    # ------------------------------------------------------
```

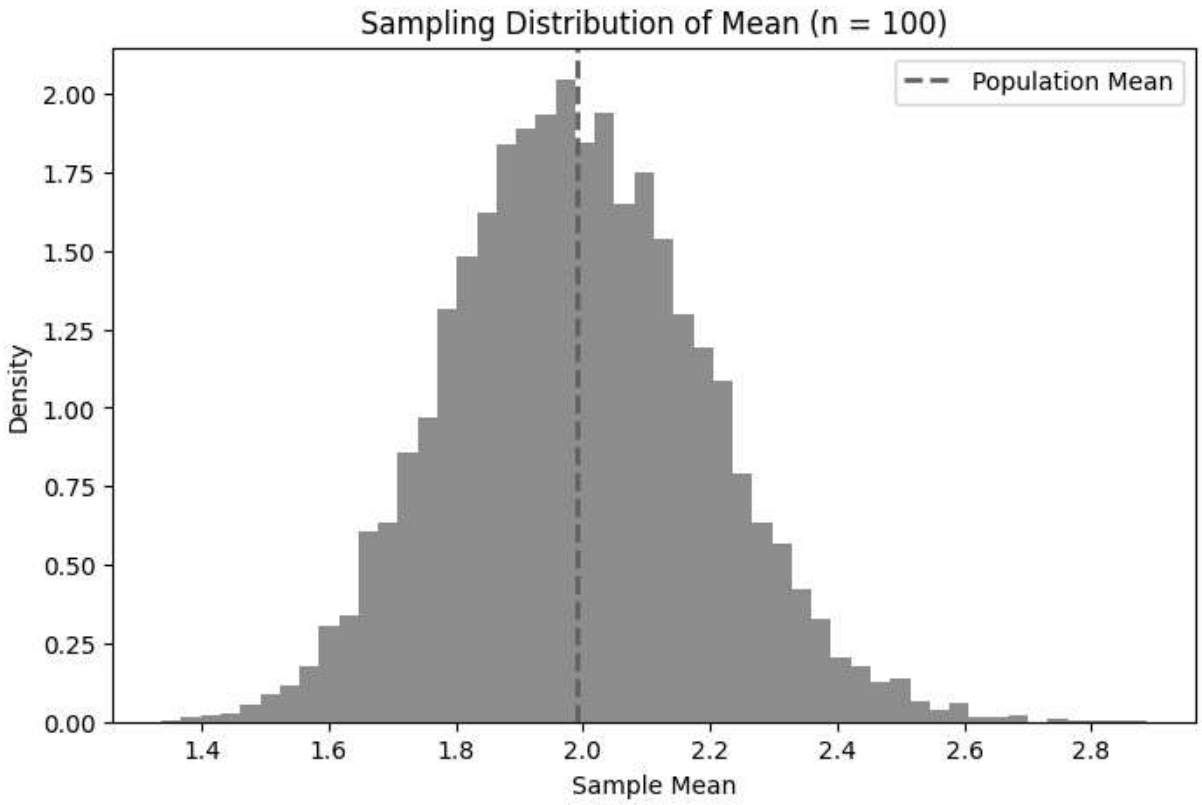
```
    # Plot sampling distribution
    # ----------------------------------------------------
    plt.figure(figsize=(8, 5))
    plt.hist(
        sample_means,
        bins=50,
        density=True,
        alpha=0.7
    )
    plt.axvline(
        population_mean,
        linestyle="--",
        linewidth=2,
        label="Population Mean"
    )
    plt.xlabel("Sample Mean")
    plt.ylabel("Density")
    plt.title(
        f"Sampling Distribution of Mean (n = {n})"
    )
    plt.legend()
    plt.show()
# ----------------------------------------------------------
# STEP 5: Display comparison table
# ----------------------------------------------------------
results_df = pd.DataFrame(results)
print("\n========== CLT SIMULATION RESULTS ==========")
display(results_df)
# ----------------------------------------------------------
# STEP 6: Compare standard errors
# ----------------------------------------------------------
results_df["SE_Difference"] = (
    results_df["Simulated_SE"]
    - results_df["Theoretical_SE"]
)
print("\n========== STANDARD ERROR COMPARISON ==========")
display(results_df[
    [
        "Sample_Size",
        "Simulated_SE",
        "Theoretical_SE",
        "SE_Difference"
    ]
])
# ----------------------------------------------------------
# STEP 7: Population distribution
# ----------------------------------------------------------
plt.figure(figsize=(8, 5))
plt.hist(
    population,
    bins=50,
    density=True
)
plt.xlabel("Population Value")
plt.ylabel("Density")
```

```
plt.title("Original Exponential Population")
plt.show()
```

```
Population Mean: 1.9919403181123696
Population Standard Deviation: 1.9859380946287923
```





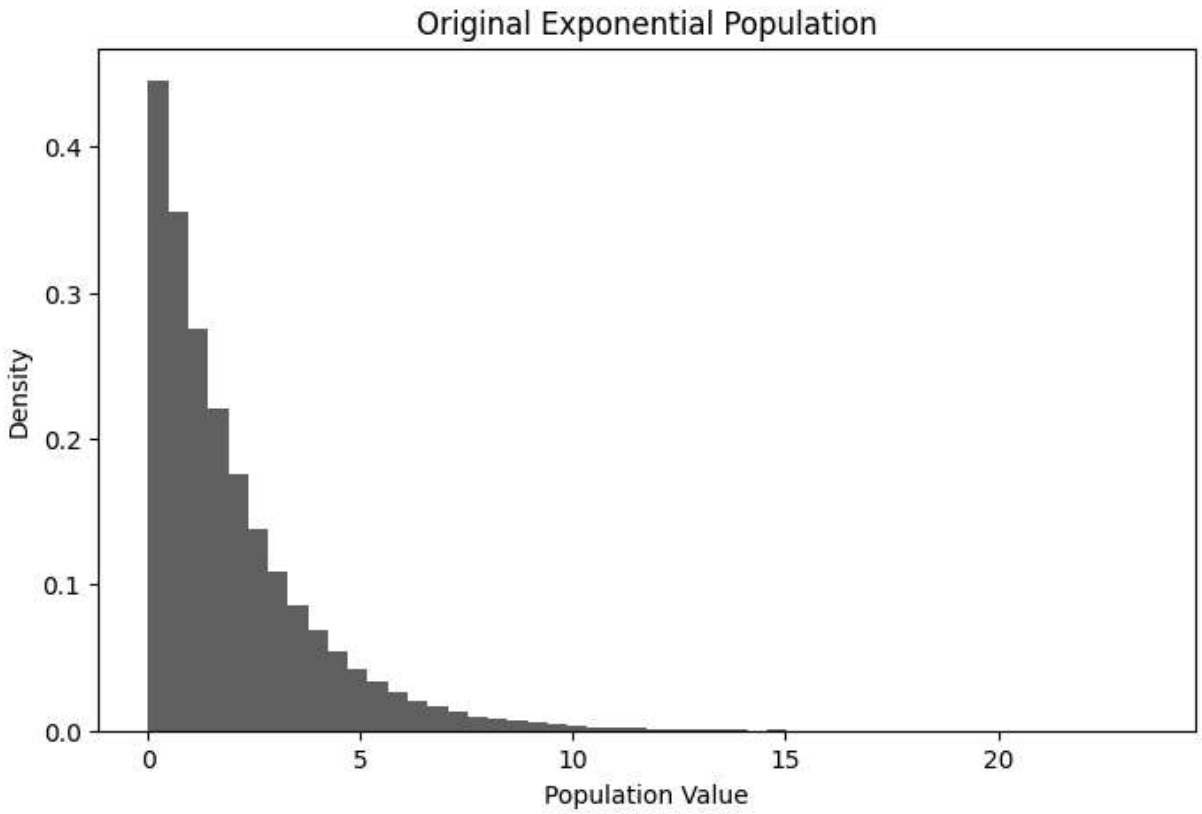


========== CLT SIMULATION RESULTS ==========

	Sample_Size	Simulated_Mean	Simulated_SE	Theoretical_SE
0	5	2.011526	0.889432	0.888139
1	30	1.995821	0.360574	0.362581
2	100	1.992211	0.201119	0.198594

========== STANDARD ERROR COMPARISON ==========

	Sample_Size	Simulated_SE	Theoretical_SE	SE_Difference
0	5	0.889432	0.888139	0.001294
1	30	0.360574	0.362581	-0.002007
2	100	0.201119	0.198594	0.002525


Original Exponential Population
Density
0.0
0.1
0.2
0.3
0.4
0
5
10
15
20
Population Value